Introduction to Software Engineering

What is Software?

The product that software professionals build and then support over the long term.
Software encompasses:
(1) instructions (computer programs) that when executed provide desired features, function, and performance;
(2) data structures that enable the programs to adequately store and manipulate information and
(3) documentation that describes the operation and use of the programs.

Dual Role of Software

Software is both a product and a vehicle for delivering a product

- **Product**
  - Delivers computing potential
  - Produces, manages, acquires, modifies, display, or transmits information

- **Vehicle**
  - Supports or directly provides system functionality
  - Controls other programs (e.g., operating systems)
  - Effects communications (e.g., networking software)
  - Helps build other software (e.g., software tools)

Software products

Generic products

• Stand-alone systems that are marketed and sold to **any customer** who wishes to buy them.
• Examples – PC software such as editing, graphics programs, project management tools; CAD software; software for specific markets such as appointments systems for dentists.

Customized products

• Software that is commissioned by **a specific customer** to meet their own needs.
• Examples – embedded control systems, air traffic control software, traffic monitoring systems.

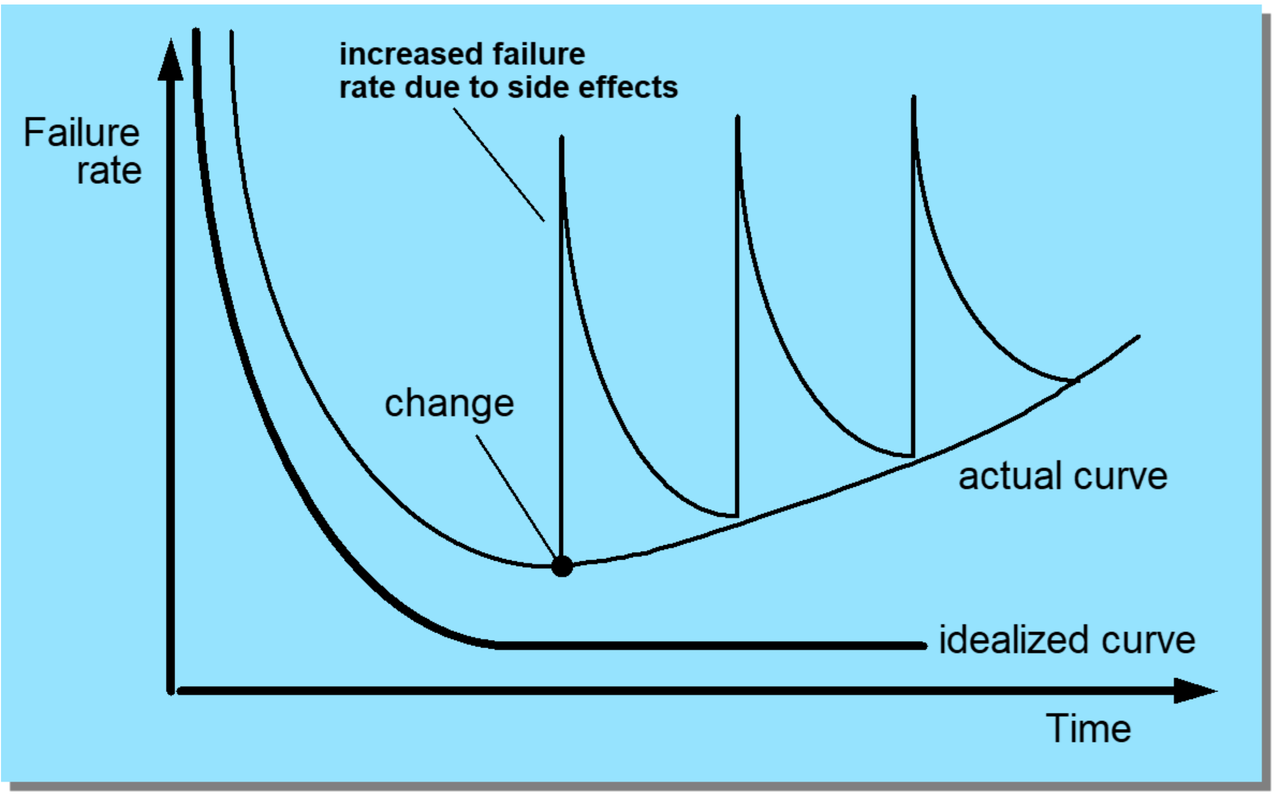
Software costs

• Software costs often dominate computer system costs. The costs of software on a PC are **often greater** than the hardware cost.
• Software costs **more to maintain** than it does to develop. For systems with a long life, maintenance costs may be several times development costs.
• Software engineering is concerned with cost-effective software development.

Features of Software?

- Its characteristics that make it different from other things human being build.
- Features of such logical system:
- Software is developed or engineered; it is not manufactured in the classical sense which has quality problem.
- Software doesn't "wear out." but it deteriorates (due to change).

- Hardware has bathtub curve of failure rate (see image below) and has a high failure rate in the beginning, then drop to steady state, then cumulative effects of dust, vibration, abuse occurs.
- Although the industry is moving toward component-based construction (e.g. standard screws and off-the-shelf integrated circuits), most software continues to be custom-built.
- Modern reusable components encapsulate data and processing into software parts to be reused by different programs. E.g. graphical user interface, window, pull-down menus in library etc.



Software Engineering Definition

The seminal definition:
[Software engineering is] the establishment and use of sound engineering principles in order to obtain economically software that is reliable and works efficiently on real machines.

The IEEE definition:
Software Engineering: (1) The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.
(2) The study of approaches as in (1).

Software Development Lifecycle

Software Development Environments